R Practical - Regression_Diagnostics

SOST10142/20142 Applied Statistics for Social Scientists

2023-2024

1 Data and variables

We will use the 2011 England Health Survey. Our dependent outcome variable is `bmival`, which stands for 'body mass indx' and is an indicator of obesity (e.g., BMI: 25-29.9 overweight, 30-39.9 obese, etc.). BMI compares an individual's weight to his or her height. Our aim is to explain variation in `bmival` while addressing potential issues caused by outliers and other model diagnostics. The main predictor variables of interest are social class and age, both of which are measured as a continuous variables. Our research question is :

> To what extent is age associated with BMI? Is young age associated with lower bmi, and how might social class impact one's BMI?

2 Learning outcome:

By the end of the practical, you should be able to identify outliers and treat them appropriately. Also, you should be able to request diagnostics of regression models and check their assumptions accordingly.

3 Prep

We first install packages and load them.

```
## Uncomment to install 'pacman' package manager (if needed)
# install.packages('pacman')

## Install the following packages:
## modelsummary         Report-ready tables
## broom                Diagnostic stats
## tidyverse            Data manipulaton
## janitor              Cross-tabulations(e.g., tabyl)
## psych                Descriptive stats table
## car                   Regression diagnostics
```

```
pacman::p_load(modelsummary, broom, tidyverse, janitor, psych, car)
```

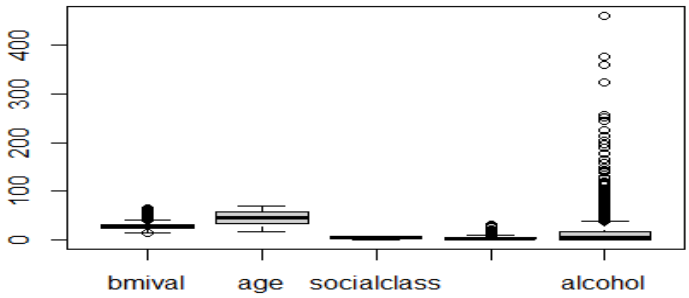
4 Identify and treat outliers

4.1 Identify outliers

We load the data and produce a boxplot to identify outliers.

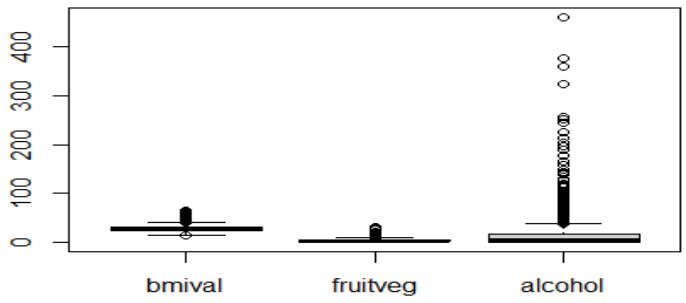
```
health <- read_rds("health_cleaned.rds")

## Visually inspect continuous measures' potential outliers
boxplotvar5 <- c("bmival", "age", "socialclass",
  "fruitveg", "alcohol")
boxplot(health[, boxplotvar5])
```



The graph shows that age and socialclass do not seem to have outliers (i.e., the circles above or below the boxes). Let's focus on the three variables that do seem to have outliers. You may also want to use a histogram to inspect the distribution of each variable (e.g., hist(health$bmival).

```
boxplotvar3 <- c("bmival", "fruitveg", "alcohol")
boxplot(health[, boxplotvar3])
```




```
hist(health$bmival)
hist(health$fruitveg)
hist(health$alcohol)
```

Clearly, these `bmival, fruitveg, alcohol` variables present outliers. To treat outliers, we first need to create an indicator variables to show which cases are outliers. Then, we can decide which option might be best, such as removing them entirely, or 'winsorising' them by replacing extreme values with a threshold of acceptable cut-off points.

In this practical, we will remove them from the analysis, but you should think carefully about whether this is a wise decision. Let's find out the highest and lowest acceptable values indicated by `mean +- 3 standard deviation (SD)`. We request a summary statistics table and inspect SD. As we will further process the output, we save the summary stats table as `outl`.

4.2 Compute upper and lower bounds

```
outl  <- psych::describe(health[, boxplotvar3], fast = TRUE)
outl

##          vars    n  mean    sd   min    max  range   se
## bmival      1 5673 27.48  5.42 13.55  65.28  51.73 0.07
## fruitveg    2 5673  3.67  2.68  0.00  30.00  30.00 0.04
## alcohol     3 5673 12.28 21.60  0.00 461.50 461.50 0.29
```

The `outl` object can be considered as `data.frame`, which is helpful.

```
class(outl)

## [1] "psych"      "describe"   "data.frame"
```

Let's remove unnecessary columns and keep columns 3 to 6 only. We also add `data.frame` explicitly to ensure `outl` can be considered as `data.frame`.

```
outl  <- data.frame(outl[, c(3:6)])
```

We then compute the upper and lower limits (bounds) by applying the formula, `mean +/- 3*sd`.

```
bounds  <- outl %>% mutate(lower = mean - 3*sd,
                           upper = mean + 3*sd)
bounds

##               mean        sd      min       max      lower    upper
## bmival   27.480455  5.417717 13.55082  65.27721  11.227302 43.73361
## fruitveg  3.666402  2.683403  0.00000  30.00000  -4.383808 11.71661
## alcohol  12.279564 21.600368  0.00000 461.50000 -52.521541 77.08067
```

You may carry out checks to see if the computation was done correctly.

```
## Check that the computation was done correctly!
27.480455 + 3*5.424481

## [1] 43.7539
```

You'll notice that all three variables' minimum values lie within the lower bound, whereas maximum values DO lie outside the upper bound of each variable. Therefore, we'll create indicators for outliers that lie beyond the upper bound.

The upper bound for each variable can be accessed by `bounds["row_name", "upper"]`. We'll use this to generate outliers later.

```
bounds["bmival",   "upper"]

## [1] 43.73361

bounds["fruitveg", "upper"]

## [1] 11.71661

bounds["alcohol",  "upper"]

## [1] 77.08067
```

4.3 Create an indicator of outliers

To create an outlier indicator for bmi, fruitveg, and alcohol, we define a criterion: if an observed value exceeds the upper bound, we assign 1 to indicate as an outlier, otherwise 0 (non-outlier). In other words, we're creating a dummy variable to indicate whether the attribute (outlier) is present. The criterion is established by `ifelse(criterion, value if true, value if false)`. See the pattern used for the other variables.

```
health  <- health %>%
 mutate(
  bmival_outl   = ifelse(bmival   > bounds["bmival","upper"], 1, 0),
  fruitveg_outl = ifelse(fruitveg > bounds["fruitveg", "upper"], 1, 0),
```

```
  alcohol_outl   = ifelse(alcohol   > bounds["alcohol", "upper"], 1, 0)
 )
```

The next step is to create a global indicator of outlier to represent outlier in any (in R, use `|` which means `or`) of the three variables. The following code instructs R to code outlier as 1 if any case contains an outlier among bmival, fruitveg, or alcohol variables. Otherwise, R records a 0.

```
health  <- health %>%
  mutate(outlier = ifelse(
    bmival_outl == 1 | fruitveg_outl == 1 |alcohol_outl == 1, 1, 0) )
```

Let's examine the extent. Can you identify how many observations are outliers in the data?

```
health %>% tabyl(outlier) %>% adorn_totals()

##  outlier    n    percent
##        0 5488 0.96738939
##        1  185 0.03261061
##    Total 5673 1.00000000
```

We can verify this with cross-tabulations.

```
health %>% tabyl(outlier, bmival_outl)
health %>% tabyl(outlier, fruitveg_outl)
health %>% tabyl(outlier, alcohol_outl)
health %>% tabyl(bmival_outl, alcohol_outl) # 1 overlapping case
```

We can verify it.

```
55 + 63 + 68 - 1 # 1 overlapping case

## [1] 185
```

4.4 Drop outliers

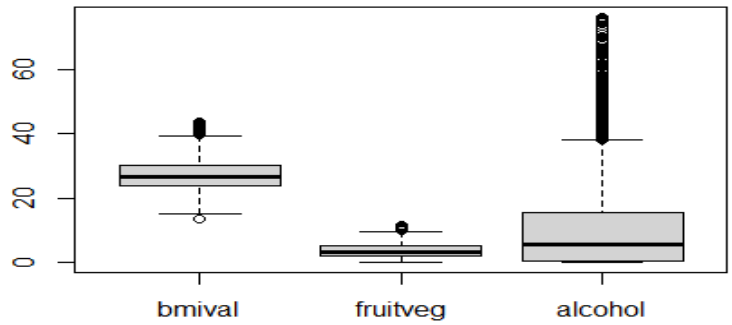
By removing outliers, we have 5488 cases left for analysis. Remember, that this is only ONE potential option, and it may not be the best course of action. You should always be transparent about your decision to treat outliers and defend this decision with reasoned arguments (i.e., explain why this is the right thing to do).

```
health  <- health %>% filter(outlier == 0)
nrow(health)

## [1] 5488
```

We also inspect using boxplots.

```
boxplot(health[, boxplotvar3])
```



It may appear that we still have outliers in the boxplots. This is because outliers in boxplots are defined by 1.5*IQR (interquartile range), which is more conservative than the z-score method we used above. As boxplots tend to pick up rather `mild` values as outliers, we can acknowledge this difference and move on to the next step.

We have just completed the process of treating univariate outliers. Yay, you! This does not necessarily mean there are no outliers present in the data. Although univariate outliers are removed, multivariate outliers may exist. These can be depicted easily with the regression model diagnostics. Before that, let's tweak our models.

5 Generate the mean-centred age

As well as removing outliers, we also change our modelling technique by using a centred value for age rather than the raw scale. This will help with interpretation of the model results because the y-intercept is determined when all predictors are set to zero. If we were to set age to 0, for example, then the y-intercept is non-sensical because a value of BMI cannot be determined for an individual that has 0 age (i.e., not even born yet).

First, print the mean age to check. Then, save the mean as a new variable, `mean_age`.

```
mean(health$age)

## [1] 45.49945

health$mean_age  <- mean(health$age)
```

The centred age is derived by `age - mean_age`. Let's reorder these age variables by relocating to the beginning of the dataset and print them for inspection.

```
health  <- health %>%
 mutate(ctr_age = age - mean_age) %>%
 select(age, mean_age, ctr_age, everything() )
```

```
head(health[, 1:3])
```

```
##   age mean_age     ctr_age
## 1  46 45.49945   0.5005466
## 2  44 45.49945  -1.4994534
## 3  48 45.49945   2.5005466
## 4  48 45.49945   2.5005466
## 5  26 45.49945 -19.4994534
## 6  43 45.49945  -2.4994534
```

6 Descriptive statistics

Inspect centred age. Also, can you see the maximum value is now bound to the upper limit after outliers are removed?

```
desc_var <- health %>%
      select(-hserial, -pserial, -gender,
             -employed)

psych::describe(desc_var, fast = TRUE)
```

```
##               vars    n  mean    sd    min   max range   se
## age              1 5488 45.50 14.17  16.00 70.00 54.00 0.19
## mean_age         2 5488 45.50  0.00  45.50 45.50  0.00 0.00
## ctr_age          3 5488  0.00 14.17 -29.50 24.50 54.00 0.19
## bmival           4 5488 27.28  5.02  13.55 43.69 30.14 0.07
## socialclass      5 5488  4.17  2.13   0.00  7.00  7.00 0.03
## fruitveg         6 5488  3.55  2.35   0.00 11.67 11.67 0.03
## alcohol          7 5488 10.83 13.98   0.00 76.48 76.48 0.19
## gender_num       8 5488  0.55  0.50   0.00  1.00  1.00 0.01
## employed_num     9 5488  0.68  0.47   0.00  1.00  1.00 0.01
## bmival_outl     10 5488  0.00  0.00   0.00  0.00  0.00 0.00
## fruitveg_outl   11 5488  0.00  0.00   0.00  0.00  0.00 0.00
## alcohol_outl    12 5488  0.00  0.00   0.00  0.00  0.00 0.00
## outlier         13 5488  0.00  0.00   0.00  0.00  0.00 0.00
```

As the dataset has changed, let's save it as a new dataset.

```
write_rds(health, file = "health_cleaned_outlierRemoved.rds")
```

7 Regression models

Let's fit several models. Model 1 uses age. Model 2 includes other predictors as control variables. For Model 3, we enter centred age instead. What changes in regression coefficients can we expect?

7.1 Baseline

```
model1 <- lm(bmival ~ age, data = health)
summary (model1)

##
## Call:
## lm(formula = bmival ~ age, data = health)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.4522  -3.4558  -0.7575   2.6493  17.4109
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 24.031361   0.223394  107.57   <2e-16 ***
## age          0.071465   0.004688   15.24   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.921 on 5486 degrees of freedom
## Multiple R-squared:  0.04064,    Adjusted R-squared:  0.04047
## F-statistic: 232.4 on 1 and 5486 DF,  p-value: < 2.2e-16
```

7.2 Age

First, fit Model 2 and record the intercept.

```
model2 <- lm(bmival ~ socialclass + age + gender +
           employed + fruitveg + alcohol, data = health)
summary (model2)  #  0.045

##
## Call:
## lm(formula = bmival ~ socialclass + age + gender + employed +
##     fruitveg + alcohol, data = health)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.7429  -3.4330  -0.8027   2.6320  17.7610
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 24.749010   0.310305  79.757  < 2e-16 ***
## socialclass -0.102744   0.032802  -3.132  0.00174 **
## age          0.074792   0.004928  15.178  < 2e-16 ***
## genderWomen -0.428079   0.137353  -3.117  0.00184 **
## employedYes  0.166113   0.153671   1.081  0.27976
## fruitveg    -0.046715   0.028976  -1.612  0.10697
## alcohol     -0.013953   0.004898  -2.849  0.00441 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 4.909 on 5481 degrees of freedom
## Multiple R-squared:  0.04605,    Adjusted R-squared:  0.045
## F-statistic: 44.09 on 6 and 5481 DF,  p-value: < 2.2e-16
```

7.3 Centred age

Next, we fit the model 3. Again, record the intercept. How do you interpret the intercept?

```
model3 <- lm(bmival ~ socialclass + ctr_age + gender +
             employed + fruitveg + alcohol, data = health)
summary (model3)  #  0.045

##
## Call:
## lm(formula = bmival ~ socialclass + ctr_age + gender + employed +
##     fruitveg + alcohol, data = health)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.7429  -3.4330  -0.8027   2.6320  17.7610
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 28.152011   0.205098 137.262  < 2e-16 ***
## socialclass -0.102744   0.032802  -3.132  0.00174 **
## ctr_age      0.074792   0.004928  15.178  < 2e-16 ***
## genderWomen -0.428079   0.137353  -3.117  0.00184 **
## employedYes  0.166113   0.153671   1.081  0.27976
## fruitveg    -0.046715   0.028976  -1.612  0.10697
## alcohol     -0.013953   0.004898  -2.849  0.00441 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.909 on 5481 degrees of freedom
## Multiple R-squared:  0.04605,    Adjusted R-squared:  0.045
## F-statistic: 44.09 on 6 and 5481 DF,  p-value: < 2.2e-16
```

7.4 Report-ready tables

We produce a summary regression table using `modelsummary` package.

```
models <- list(model1, model2, model3)

modelsummary(models, stars = TRUE,
             gof_map = c("nobs", "r.squared", "adj.r.squared" ))
```

	Model 1	Model 2	Model 3
(Intercept)	24.031***	24.749***	28.152***
	(0.223)	(0.310)	(0.205)

	Model 1	Model 2	Model 3
age	0.071***	0.075***	
	(0.005)	(0.005)	
socialclass		-0.103**	-0.103**
		(0.033)	(0.033)
genderWomen		-0.428**	-0.428**
		(0.137)	(0.137)
employedYes		0.166	0.166
		(0.154)	(0.154)
fruitveg		-0.047	-0.047
		(0.029)	(0.029)
alcohol		-0.014**	-0.014**
		(0.005)	(0.005)
ctr_age			0.075***
			(0.005)
Num.Obs.	5488	5488	5488
R2	0.041	0.046	0.046
R2 Adj.	0.040	0.045	0.045

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

7.5 F-test as a model comparison

Model 2 and 3 are essentially identical models in terms of model fit index. We thus can compare the simple model (model 1, baseline) and the complex model (model 3). Is model 3 superior to model 1? The ANOVA results will tell us (i.e., a statistically significant F-test suggests Model 3 explains more variation than Model 1).

```
## anova(model1)
anova(model1, model3)

## Analysis of Variance Table
##
## Model 1: bmival ~ age
## Model 2: bmival ~ socialclass + ctr_age + gender + employed + fruitveg +
##     alcohol
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1   5486 132853
## 2   5481 132104  5    748.05 6.2073 9.55e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

8 Diagnostics

We need to check assumptions made in the regression by requesting diagnostic graphs. The visual inspection is required to confirm whether the assumptions are met[1].

8.1 Multicollinearity

Some explanatory variables are highly correlated, leading to multicollinearity. You may detect this in a correlation matrix. We can also request `vif` to examine multicollinearity after regression models are fitted. VIF stands for 'variance inflation factor'. Let's use `cbind` to print vif in one column. Are all variables within the acceptable cut-off of VIF, say 10?

```
## VIF for Model 3
cbind(vif(model3))

##                 [,1]
## socialclass 1.114717
## ctr_age     1.110246
## gender      1.062584
## employed    1.178690
## fruitveg    1.057267
## alcohol     1.067432
```

8.2 Homoscedasticity - constant error variance

We can perform a formal diagnostic test for homoscedasticity. The test assumes homoscedasticity. As such, we expect the test result to be non-significant. By contrast, if the test result is significant ($p<0.05$), the assumption is violated, and we can conclude that there is evidence of non-constant variance. This may be the case when data is clustered (e.g., individuals nested in families, schools, etc). For clustered data, we need more advanced modelling techniques (beyond the scope of the course!). Is the homoscedasticity assumption of linear regression met?

```
## Cook/Weisberg constant error variance test
ncvTest(model3)

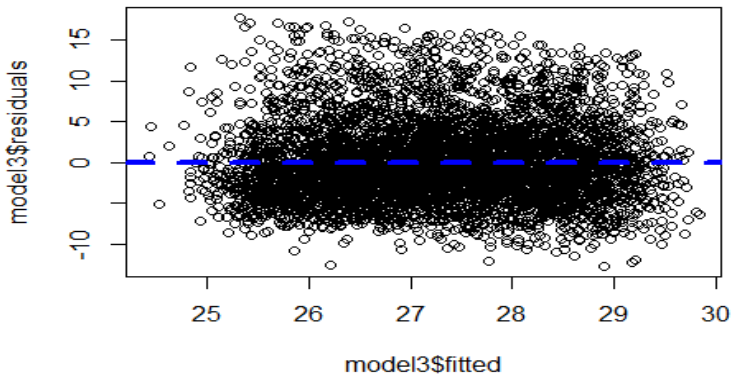
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 0.02574654, Df = 1, p = 0.87252
```

[1] There are other approaches for diagnostics, and Kabacoff (2015) offers a comprehensive overview in his book, R in Action (pp.182-201). The test criteria can be arbitrary by authors. I used the criteria that deemed most common in literature.

8.3 Linearity

The regression models assume a linear relationship between the dependent variable and independent variables. If this assumption is met, no systematic relationship is seen between residuals and predicted values (e.g., we do not expect a curved line). We can visualise this by plotting predicted values against residuals.

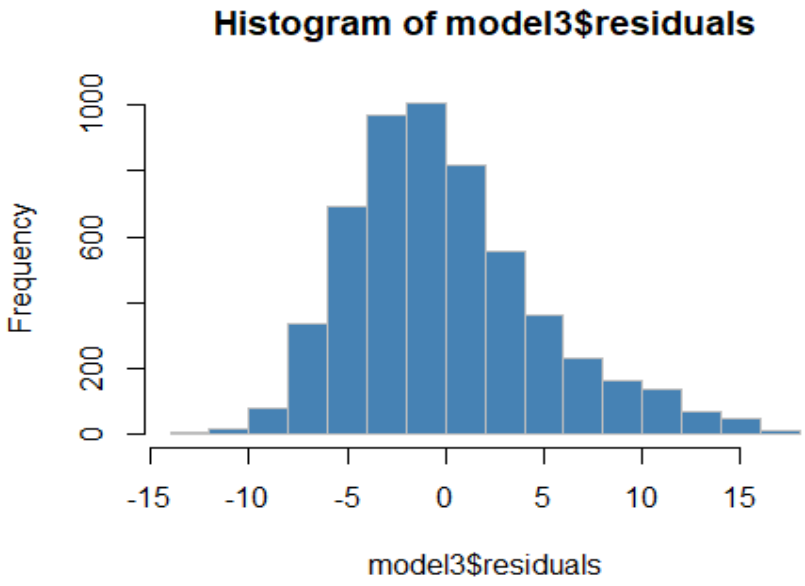
```
plot(x = model3$fitted, y = model3$residuals)
abline(h = 0, lty = 2, lwd = 4, col = "blue")
```



8.4 Normality

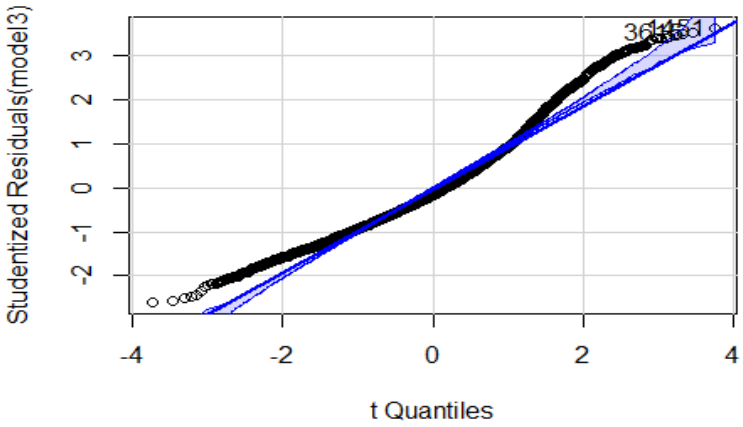
Let's visualise residuals. Do the residuals follow a normal distribution well?

```
hist(model3$residuals,
  col = "steelblue", border = "grey")
```



As we observe a wide range of (raw) residuals, it is not straightforward to quantify which cases are considered as a departure from normality. The range will vary by data, as well. Therefore, we need some kind of `standarised` plot. The normal Q-Q plot is a probability plot of the standardised residuals against the values that would be expected under normality (Kabacoff 2015, p. 183). If the model meets the normality assumption, the points on the graph should fall on the straight 45-degree line. We can also identify cases lying outside of normal distribution (e.g., +/-3 SD of residuals). What does the printed case number suggest?

```
qqPlot(model3)
```




```
## [1] 1451 3615
```

Despite some potential (multivariate) outliers, the model seems to meet the normality assumption fairly well. Let's look at other diagnostics before we decide how to treat these issues.

8.5 Outliers and influential observations

After fitting regression models, R stores diagnostic data. To access this information, we need a package called, `broom`. To request diagnostics statistics, we use this function from the package: `augment(regression_model)`.

```
model3aug  <- augment(model3)
# View(model3aug)
```

We already know about fitted values and residuals. We also see hat values, cook distance, and studentised residuals. Influential observations may greatly 'influence' model parameters (i.e., regression coefficients); hence, the name. It is thus important to identify potential influential observations by using measures of `leverage` and `Cooks's distance`.

8.5.1 Outliers

As the studentised residuals follow a *t*-distribution, observations outside +/-2 SD range are (potential) `outliers`. Do we need to remove them? Maybe. Let's see how influential these are first.

8.5.2 Leverage

An observation with high leverage may be considered an outlier. The common measure of leverage is a hat value. The diagonal element of the hat matrix, which is a standardised measure of the distance between x values for a participant and the means of the x-values for all participants. This can be technical and it is not essential to understand hat matrix. The average hat values can be obtained by (k + 1)/n, where k is the number of explanatory variables. If cases' hat values exceed twice the average hat values, they are viewed as high leverage points.

```
mean(model3aug$.hat)

## [1] 0.00127551

avg_hat  <- 7 / 5488   # (k+1)/n
avg_hat   # 0.00127551

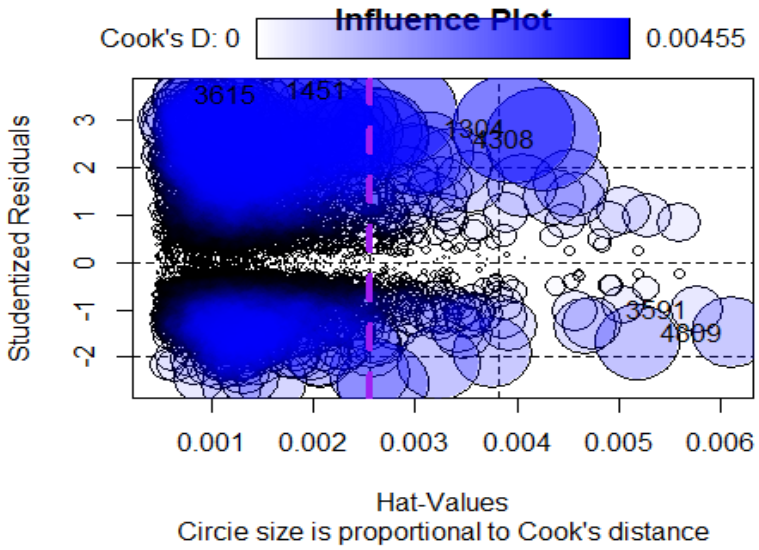
## [1] 0.00127551
```

Let's plot hat values and draw the cut-off line defined by `avg_hat*2`.

```
library(car)
influencePlot(model3, main = "Influence Plot",
     sub = "Circie size is proportional to Cook's distance")
```

```
##         StudRes          Hat        CookD
## 1304  2.8261376 0.0039757827 0.0045487077
## 1451  3.6247481 0.0016552671 0.0031051630
## 3591 -0.9964887 0.0057513609 0.0008205836
## 3615  3.5148273 0.0007635752 0.0013458443
## 4308  2.5853638 0.0042554732 0.0040765715
## 4809 -1.4812224 0.0060926482 0.0019209148

abline(v = avg_hat*2, lty = 2, lwd = 4, col = "purple")
```



In the plot, to the right of purple line exhibits high leverage points. We see that observation 1304 lies outside the +/-2 range of studentised residuals and displays a high leverage. Also, have a look at the table printed.

Although the graph offers some visualisation, these two measures are not combined in one, and without a cut-off criterion, high leverage cases do not necessarily mean influential. Cook's distance uses both studentised residuals and hat values to offer a measure of influential observations.

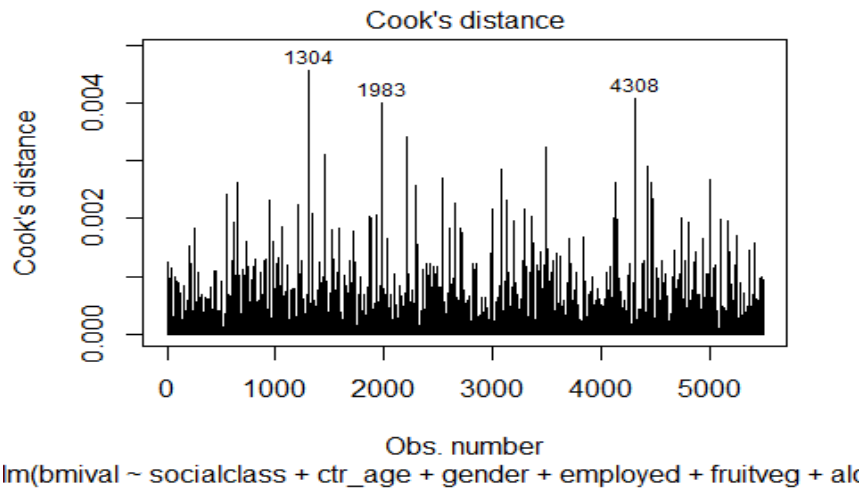
8.5.3 Cook's distance

Cook's distance indicates whether observations have any undue influence on the regression model results. Values of Cook's distance of 1 or greater are considered problematic (Tabachnick & Fidell, 2007, p.75)[2]. If Cook's distance is above 1, you may need to consider removing the case(s). But remember, this decision can create other problems, so proceed with caution!

[2] Tabachnick, B. G. & Fidell, L. S. (2007). Using multivariate statistics (5th edn). Boston: Pearson Education.

R produces several diagnostic plots for regression models. To request Cook's distance plot, we specify `which = 4`.

```
plot(model3, which = 4)
```



Let's check the maximum value. Is there any case(s) that appears influential?

```
options(scipen = 999)  # For non-scientific notation
summary(model3aug$.cooksd)

##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## 0.00000000 0.00001498 0.00006604 0.00018260 0.00020270 0.00454871
```

9 Removing cases

If you decide to remove cases(observations), you may try something like this:

```
drop   <- c(1451)
health <- health[-c(drop), ]
```

10 Conclusion

What can you conclude? If you identify potential outliers, you may remove the case and rerun the models. In such cases, it is likely that R identifies other potential outliers. If you do, pay close attention to the R-squared values and regression paramters changes. If there is no improvement, you may return to the older model and conclude the analysis.

In empirical research, it is researcher's responsibility to make reasonable decisions! You need to be able to justify your decisions. Whichever decision you make, be cautions and ensure that model fits the data, not the other way!